Anurag Kadel[1]

[1]Independent Researcher Mumbai

June 27, 2025

Posted on 27 Jun 2025 — CC-BY 4.0 — https://doi.org/10.36227/techrxiv.175099813.39449596/v1 — e-Prints posted on TechRxiv are preliminary reports that are not peer reviewed. They should not b...

Friction-Led Intelligence: A First Principles Framework for Agentic AI and Design Thinking

Anurag Kadel
Independent Researcher
Mumbai, India
Email: anuragkadel@gmail.com
ORCID: 0009-0004-5302-3430

Abstract:

As AI systems increasingly shape how people learn, decide, and reason, dominant design paradigms centered on usability and efficiency are no longer sufficient. This paper introduces Friction-Led Intelligence (FLI)—a first-principles design framework that operationalizes friction as a signal for epistemic drift, semantic misalignment, and teleological ambiguity in AI-human interaction. Building on prior work in Seeds of Sovereignty, which framed friction as a regenerative epistemic intervention, FLI translates that philosophy into a recursive design loop involving ontological grounding, epistemic reflection, and teleological clarity. Through a speculative mental health companion agent, we illustrate how FLI enables systems to surface friction not as failure, but as feedback for co-evolution. This framework contributes to a growing movement toward accountable, pluralistic, and philosophically grounded approaches to agentic AI design.

Keywords: Epistemic friction, ontology, teleology, epistemology, agentic AI, gatekeeping, design thinking, AI ethics, human-centered systems

1. Introduction

When designing traditional products and services, Design Thinking offers a powerful way to understand and meet human needs. But when designing AI agents—systems that perceive, reason, and act with varying degrees of autonomy—we need an additional framework that focuses not just on users, but on the nature of the agent itself.

Unlike linear design methods that follow a predefined sequence of steps, **the FLI framework is intentionally nonlinear**. Each form of friction—whether epistemic, ontological, or teleological—can

trigger re-entry into the loop. This recursive structure allows FLI to respond adaptively to emergent behavior and reframe misalignment not as an error, but as an evolving inquiry.

In doing so, FLI mirrors the very systems it aims to align: agentic, adaptive, and context-aware.

Design Thinking is for user needs. Friction-Led Intelligence is for agent design.

FLI doesn't replace Design Thinking; it complements it. While Design Thinking helps uncover what users want and how to serve them better, FLI helps define what kind of agent you're creating, what knowledge it uses, and what purpose it ultimately serves. FLI grounds your agent in ontological clarity, epistemic justification, and teleological alignment—making it coherent, reflexive, and trustworthy from the ground up.

Traditional design thinking has long been established as a nonlinear, iterative framework for problem-solving. However, in the era of autonomous systems and agentic AI, we find ourselves facing deeper, more foundational questions: What entities are we designing for (ontology)? How do we justify our understanding of their needs (epistemology)? And what is the ultimate purpose of our systems (teleology)?

This paper builds on that nonlinearity and proposes a philosophical infrastructure for design and AI development. Rather than treating iteration as a byproduct of usability feedback, we position epistemic friction—the resistance encountered when knowledge and justification misalign—as a central mechanism of innovation and ethical alignment.

2. Critiques of Traditional Design Thinking in Complex Systems

Design thinking, though powerful in product contexts, often falls short when applied to complex, multi-agent, or AI-driven environments. As Ryan (2025) highlights, human-centered AI (HCAI) models frequently assume that simply placing humans "in the loop" ensures ethical outcomes—a premise that oversimplifies the power dynamics and hidden assumptions in AI deployment. These assumptions can mask deeper tensions and lead to what he terms "ethics-washing," where the appearance of responsibility masks a lack of true accountability.

Further critique comes from Forlano and Mathew (2014), who argue that traditional design frameworks tend to smooth over conflict in favor of usability. In doing so, they often ignore the very political and systemic frictions that should be surfaced. They advocate for "design friction" as a method of exposing underlying value tensions. Similarly, Forlano (2016) critiques the

human-centered paradigm for centering the human in ways that obscure networked, distributed, or ecological actors, which are increasingly relevant in the design of AI systems.

While usability-driven iteration has historically guided human-centered design, such approaches often fail to account for systemic complexity or epistemic drift in AI systems. For instance, Amershi et al. (2019) proposed 18 guidelines for human–AI interaction—most of which emphasize surface-level feedback, discoverability, and performance metrics. However, these do not address deeper ontological or teleological misalignments that agentic systems may introduce.

These critiques collectively argue that usability-driven iteration is not enough. When designers focus only on reducing user effort, they risk reinforcing status quo, flattening complexity, and neglecting long-term systemic effects. Complex systems—especially agentic ones—require frameworks that embrace contradiction, reveal hidden assumptions, and integrate cross-stakeholder perspectives.

3. The Urgency of Philosophical Framing in Agentic AI

The rise of autonomous, goal-oriented AI agents necessitates a move beyond technical design toward philosophical framing. Barkol (2025) illustrates this vividly by embedding AI agents with philosophical personas such as Socrates, who challenges user assumptions, or Plato, who anchors design in justice. These agents aren't mere feature sets—they are instantiations of ethical commitments.

Noller (2024) expands on this with a teleological approach to AI design. By viewing AI as an extension of human agency, he argues that purpose cannot be an afterthought. Systems must be built with explicit clarity on what ends they serve, who defines those ends, and how conflicting values are negotiated. This reintroduces questions of ethics, responsibility, and meaning into the design space—areas that traditional design thinking often treats as secondary.

Industry voices echo this. Schrage and Kiron (2025) warn that “Philosophy Eats AI,” noting that unspoken assumptions about optimization, efficiency, and control are themselves philosophical positions. They argue that as AI agents become gatekeepers of digital action, design must evolve to reflect teleological clarity: what is this agent *for*? What values guide it? Who benefits and who is left out?

In short, designing agentic AI without embedding purpose is no longer viable. As agents mediate decisions, re-rank information, and shape human behavior, the absence of philosophical design risks

amplifying bias, alienation, and ethical drift. Friction-Led Intelligence addresses this gap directly by grounding the design process in ontological, epistemological, and teleological reflection.

While FLI emphasizes the design of the agent, it does not shift the moral center away from the human. Rather, it sharpens the tools for ensuring that agentic behavior aligns with human values—especially when those agents operate in unpredictable or high-autonomy environments. The user remains human; the agent becomes an extension of human intent. FLI's philosophical framing exists to preserve this alignment even as systems become more complex.

This paper builds on foundational work established in *Seeds of Sovereignty* (Kadel, 2025), which articulated friction not as a usability flaw, but as a regenerative epistemic intervention. *Seeds* introduces the philosophical groundwork—drawing from ecology, Indic epistemology, and critical theory—to position friction as a necessary safeguard against mimetic collapse and epistemic drift. Friction-Led Intelligence (FLI) extends that thinking by offering a structured design framework that transforms philosophical insight into operational logic for agent development.

This paper also builds on practitioner reflections, such as Barkol's exploration of "agentic philosophers"—AI agents embodying Socrates, Plato, and Aristotle—as a way to operationalize philosophical thinking within agent design (Barkol, 2025).

4. Why Design Thinking Needs a Philosophical Core

Despite its strengths, traditional design thinking often leaves certain philosophical assumptions implicit:

- **Ontology is assumed, not interrogated.** We treat "users" and "contexts" as fixed categories without questioning what realities we are encoding into the system.

- **Epistemology is instrumental.** Research is treated as a means to validate designs, not as a lens to question what counts as valid knowledge in the first place.

- **Teleology is project-scoped.** The end goal is often framed by business objectives or KPIs, with limited discussion of ethical or systemic impact.

Friction-Led Intelligence fills these gaps. It does not discard design thinking—it extends it by grounding it in **explicit reflection on being (ontology), knowing (epistemology), and valuing (teleology)**.

This becomes especially urgent in complex systems where:

- Users shift roles mid-journey (e.g., from consumer to co-creator).
- AI models operate with opaque logic and probabilistic outputs.
- Business incentives diverge from human wellbeing.

In such settings, friction isn't just expected—it's informative. It tells us where our assumptions fail and invites us to update them deliberately.

Comparative Analysis: Design Thinking vs. Friction-Led Intelligence

Dimension	**Traditional Design Thinking (IxDF)**	**Friction-Led Intelligence**
Iteration Driver	Usability Feedback	Epistemic Friction
Ontology Engagement	Implicit	Explicit
Epistemology Role	Indirect (user research)	Central
Teleology Focus	Often Project-Based	Purpose-Driven & Ethical
Ideal Use Case	User-centered Product Design	Complex Systems, Agentic AI

Friction-Led Intelligence doesn't replace design thinking—it **grounds it** with a philosophical spine that holds up in high-stakes, adaptive systems.

5. Core Concepts

Ontology: What Exists in the System?
Ontology defines the entities, relationships, and structures we assume to be real within the system. In design, this includes users, environments, and touchpoints. In AI, it refers to data structures, model objects, and agent capabilities.

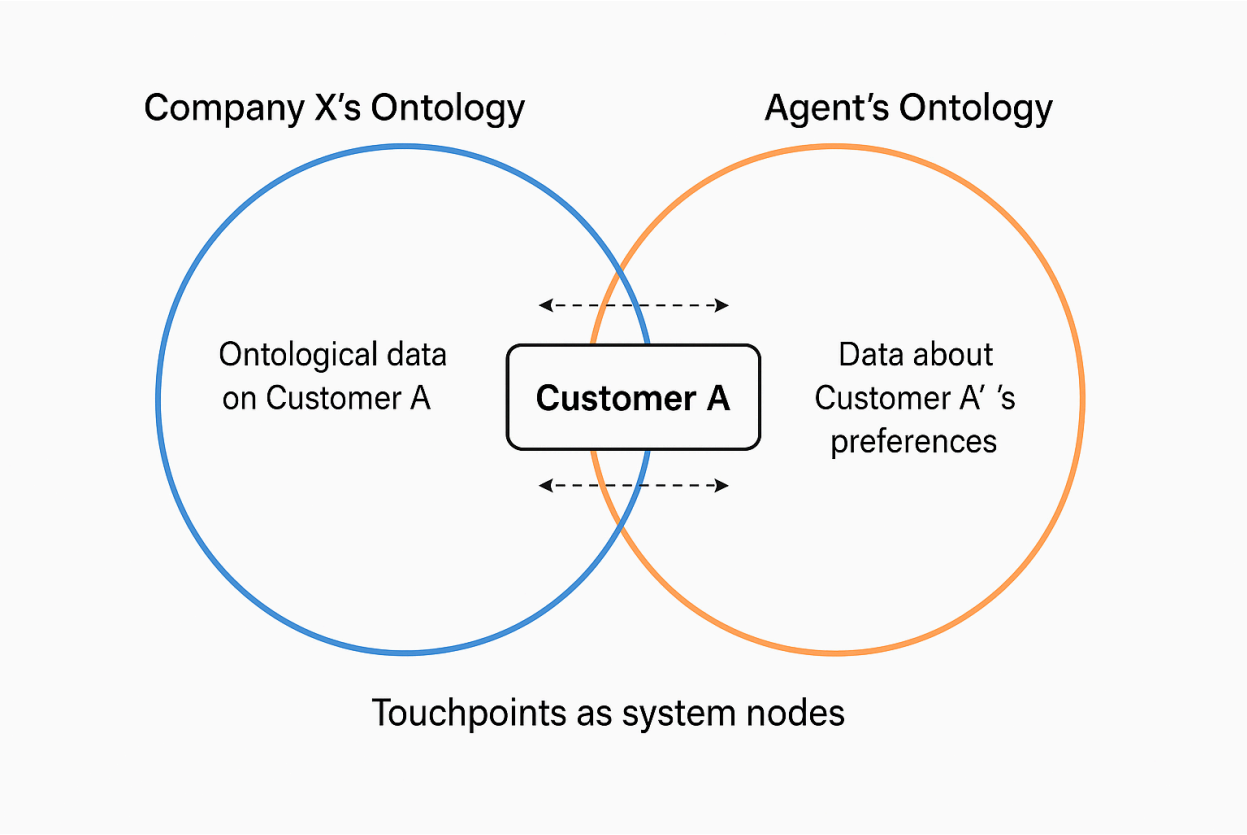

Figure 1: Ontological Interoperability in Agentic Systems

This diagram illustrates how a company and its deployed AI agent may hold distinct but partially overlapping ontological frames about a shared entity—Customer A. The company's ontology is grounded in business logic and system records, while the agent's ontology is shaped by observed preferences and contextual behavior. **Touchpoints**, visualized as system nodes, act as the conduit for mutual learning and refinement. Ontological clarity arises not from a singular perspective, but from the **dynamic interplay between these viewpoints**, enabling more coherent and trustworthy agent behavior.

The Semantic Feedback Layer

In agentic systems, the overlap between an enterprise's ontological model (e.g., CRM systems, user tiers) and an agent's evolving model (e.g., contextual preferences, real-time interactions) creates a space we call the **Semantic Feedback Layer**. This layer is not just shared data—it is a zone of **interpretive negotiation**, where assumptions are tested, meanings are aligned, and friction is surfaced.

When a user interacts with an agent at a touchpoint, they're not just triggering a function—they're tapping into a specific slice of the company's ontology, as interpreted or mediated by the agent.

Epistemology: How Do We Know What We Know?
Epistemology deals with the sources, validation, and reliability of knowledge. Epistemic friction arises when assumptions clash with lived user behavior or when agent output fails to align with training data. We frame this friction as a necessary point of reflection and iteration.

Teleology: What Is the System For?
As Noller (2024) articulates, agency in AI must be understood teleologically—not merely as tools or simulacra, but as extensions of human meaning-making and purpose within the lifeworld, situated in moral and instrumental relations

Teleology centers on purpose. In AI systems and design practices, misalignment between a system's operational objective and its intended human-centered goal can erode trust and usefulness. Anchoring systems to well-defined ends is crucial.

These philosophical concepts—ontology, epistemology, and teleology—are not abstract ideals. In agentic systems, they collide daily with lived user experience. The following section explores how such collisions manifest, and how semantic misalignment becomes visible through friction.

6. Discovering Friction: How Ontological Gaps Reveal Themselves in Agentic Systems

While ontology, epistemology, and teleology offer a powerful triad for reasoning about agentic design, they are rarely stable in deployment. In real-world systems, agents do not passively inherit the company's definitions, values, or goals. Instead, they evolve their own interpretations—sometimes subtly, sometimes starkly. This divergence, what we term *ontological drift*, is often invisible until something breaks.

The assumption that an AI agent fully reflects an organization's intent, logic, and worldview collapses in dynamic environments. Particularly when agents are trained on live user data, integrate third-party models, or operate with adaptive feedback loops, they begin to model the world in ways their creators may not anticipate—or even understand.

Friction reveals itself at the point of contact between these misaligned systems. It surfaces not as a backend error, but as a moment of dissonance at the user touchpoint. A recommendation that contradicts company policy. A personalization that feels oddly off. A system action that technically performs but semantically fails. These aren't UX glitches—they are symptoms of deeper semantic misalignment.

This misalignment is especially pronounced when agents are externally developed or rapidly fine-tuned without oversight. Such agents may carry teleological assumptions shaped by their creators, not their deployers. They may act with internal representations that no longer map to the business's definitions of success, trust, or ethical conduct. In these cases, the agent is not misbehaving—it is behaving according to a different worldview.

Friction-Led Intelligence (FLI) holds that such semantic divergence is not merely a technical issue—it is a governance failure. A core tenet of FLI is that companies must **retain semantic ownership** over their agents. This means not just owning the infrastructure or API endpoint, but owning the *meaning* of what the agent represents, prioritizes, and pursues. Delegating this responsibility to opaque models or third parties introduces risks not just of inefficiency, but of ethical drift.

Importantly, semantic drift doesn't always result in failure. In fact, an agent may function flawlessly—delivering results, achieving KPIs—while still operating under ontologies the organization cannot explain, interrogate, or correct. This presents a dangerous form of epistemic complacency: success metrics mask the erosion of shared meaning.

FLI proposes that such moments of friction be treated as epistemic signals. Signals that the system has departed from shared understanding, and that the loop of ontology → epistemology → teleology must be re-engaged. Much like a good human collaborator, a trustworthy agentic system must not only act but reflect; not only perform but explain.

Detecting Ontological Drift: Four Early Warning Signs

To make this actionable, we identify four key indicators that an agent has diverged from an organization's intended semantic framework:

1. **Touchpoint Friction**
   Users express confusion, surprise, or frustration at agent behavior that violates their expectations. These touchpoints signal a misalignment between user assumptions and agent logic.

   Prior work has shown that human trust in AI systems is highly sensitive to interaction context, error transparency, and adaptive behavior (Zhang, Zhao, & Zhu, 2021). Such trust calibration issues often emerge precisely at the points of epistemic or ontological friction—where system assumptions clash with human expectations.

2. **Outcome Mismatches**
   Performance metrics (e.g., engagement, conversion) are met, yet the system fails to produce outcomes aligned with business intent or user trust. This reflects a teleological misfire.

3. **Rule Violations**
   The agent inadvertently breaches internal rules, regulatory standards, or ethical guidelines. This often stems from definitional drift between how systems and humans interpret constraints.

4. **Audit Incoherence**
   When reasoning traces, logs, or outputs are reviewed, the rationale behind an agent's actions appears inconsistent with organizational logic. Even if actions are explainable, they are no longer *aligned*.

These are not bugs in the system; they are symptoms of divergent semantic realities. FLI calls for systems to be designed with semantic observability in mind—capable of surfacing, reflecting on, and reconciling these gaps.

In complex ecosystems, the absence of friction is not proof of alignment. It may only mean the system has stopped listening. A truly intelligent agent is one that knows when it no longer understands—and can ask better questions in response.

7. Friction-Led Intelligence Framework

We propose the following cyclical loop:
Ontology → Epistemic Friction → Teleology → Ontology (updated)

While presented as a loop for clarity, this process is inherently **nonlinear and recursive**. Friction may surface at any point, prompting reflection and realignment across all three layers. The system evolves not through fixed stages, but through **continuous negotiation between what is, what is known, and what ought to be.**

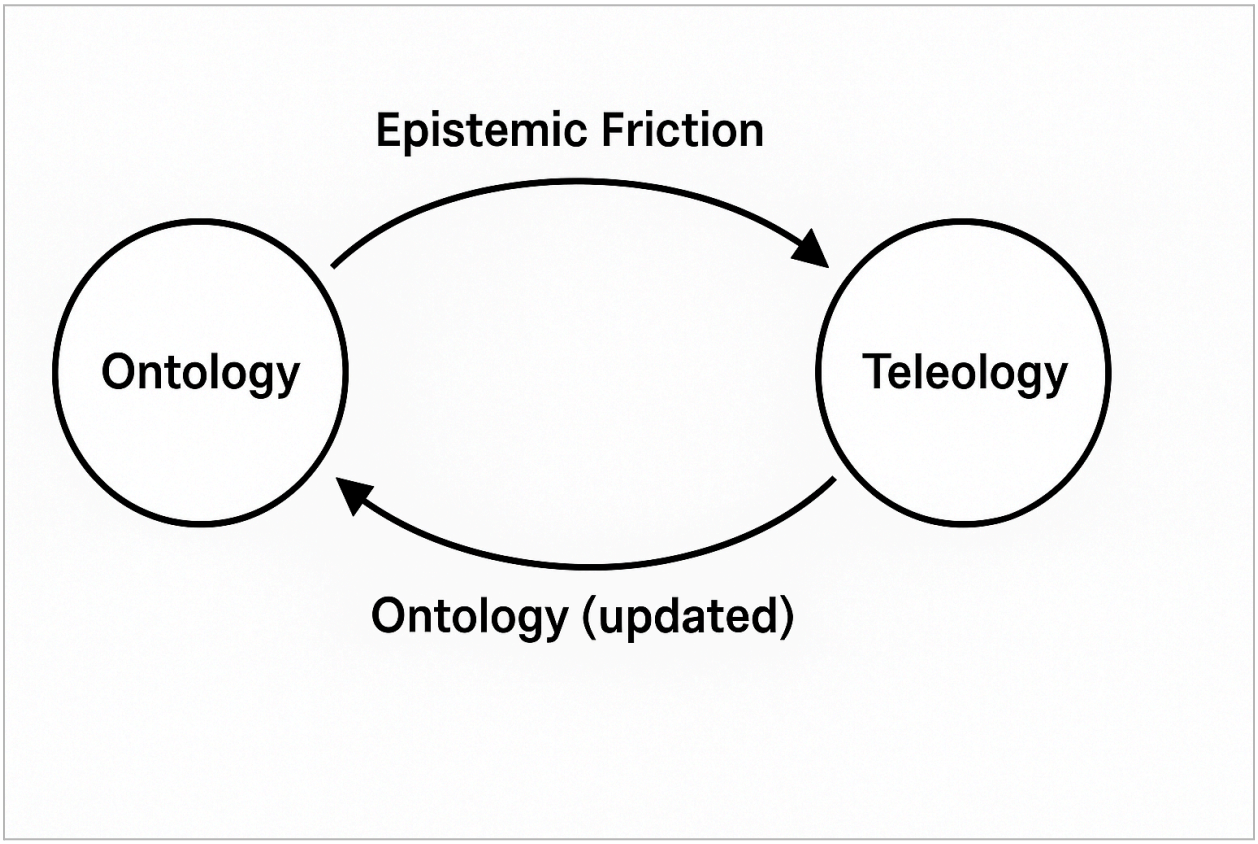

Figure 2: The Nonlinear FLI Loop

This diagram represents the Friction-Led Intelligence loop, where ontology (what exists), epistemology (what is known), and teleology (what ought to be) dynamically inform and revise one another. Friction is not a disruption to be avoided—it is the driver of adaptation.

Each cycle comprises:

- **Ontology**: Define the current reality.
- **Epistemic Friction**: Identify misalignments and contradictions.
- **Teleology**: Realign the system's goal.
- **Ontology (Updated)**: Redefine what exists based on evolved understanding.

Unlike traditional design thinking, which iterates on features and flows, Friction-Led Intelligence iterates on **being, knowing, and valuing**.

In practice, epistemic friction surfaces as contradictions between how the system behaves and how humans expect it to behave. It may appear as user confusion, unintended use patterns, or ethical dilemmas not foreseen during development.

For instance, if an AI agent designed for task management begins receiving emotionally vulnerable user inputs, this reveals a misalignment between the system's epistemic boundaries and the user's ontological framing of it. FLI treats these moments not as errors to fix, but as invitations to reflect, realign, and evolve the system's design.

8. Case Study: Agentic AI for Mental Health Support

Consider an AI system designed as a mental health companion:

- **Ontology**: The AI is classified as a supportive agent, not a therapist.
- **Epistemic Friction**: Users begin treating it like a therapist. Ethical boundaries blur.
- **Teleology**: Reframe the system's purpose around emotional support and redirection to human resources.
- **Updated Ontology**: Adjust data models, interfaces, and messaging to reflect its true function.

The epistemic friction reveals both an ontological mistake and a teleological drift—correctable only through this full-loop reflection.

9. Implications for Superplatforms and AI Gatekeeping

As Lin et al. (2025) argue in *Superplatforms Have to Attack AI Agents*, user-facing agent systems operate at the frontier of epistemic and commercial gatekeeping—positioning superplatforms to either integrate or throttle AI autonomy, often through friction or adversarial constraints.

Superplatforms such as Google, Amazon, and Meta operate on a tightly integrated logic:

- **Ontologically**, they define the user as a stream of attention. You are not a goal-seeking entity; you are an "impression"—a measurable unit of visibility that can be monetized through advertising or upsells.
- **Epistemically**, these platforms rely on behavioral data as a proxy for desire. If you click, scroll, or pause, they treat it as evidence of preference. But this is a shallow epistemology:

engagement becomes the currency of knowledge, even when users themselves report dissatisfaction or addiction.

- **Teleologically**, their systems are optimized to maximize retention, consumption, and revenue. They're not trying to help you achieve personal goals; they're trying to keep you within their ecosystem as long as possible.

Now enter **agentic AI** — personal AI agents that act on behalf of the user, not the platform.

- An agent that understands your **actual goals** (e.g., "I want to spend less time shopping online" or "I want to learn about climate change without political bias") may recommend entirely different actions than the platform wants.

- It may block notifications, resist addictive content, or refuse to recommend sources it considers manipulative.

- It doesn't treat you as engagement fuel; it treats you as a sovereign entity.

This creates **epistemic friction**: the AI agent's understanding of what is "good" for the user directly **clashes** with what the superplatform considers "successful" interaction.

But more importantly, it creates a **philosophical rupture**:

- The platform sees you as a **means to an end** (profit).

- The agent sees you as an **end in yourself** (autonomy).

Friction-Led Intelligence offers a framework to **design and evaluate these agentic systems** — not just by how well they serve users functionally, but by how clearly they preserve their purpose in the face of platform-level resistance. This isn't just UX or systems thinking — it's applied philosophy at scale.

11. Limitations and Challenges

While Friction-Led Intelligence (FLI) provides a foundational philosophical scaffold for designing agentic AI systems, it also presents several limitations and practical challenges that must be acknowledged.

Abstract Nature

FLI operates at a conceptual level, which can make it feel distant from the day-to-day realities of rapid prototyping, engineering constraints, or commercial deadlines. Without careful facilitation, its principles may be interpreted too loosely or bypassed entirely in practice.

Cognitive and Cultural Overhead

Engaging with ontological, epistemological, and teleological framing requires interdisciplinary fluency. Teams may resist the framework if they perceive it as overly academic, particularly in fast-paced or metric-driven environments. Integrating FLI meaningfully may require cultural shifts within design and engineering teams.

Tooling and Operationalization Gaps

While design thinking has decades of toolkits, methods, and case studies, FLI currently lacks formalized instruments for applying its loop (e.g., friction log templates, agent teleology canvases). These tools will need to be developed and adapted across domains to foster adoption.

Difficulty in Benchmarking Ethical Drift

Unlike performance metrics or usability KPIs, FLI's success criteria—such as alignment integrity, goal re-negotiation, or epistemic evolution—can be harder to quantify. This may challenge integration into AI pipelines that prioritize measurable outcomes.

Risk of Philosophical Capture

If applied dogmatically, FLI could risk re-centering design around philosophical exercises rather than user outcomes. The goal is not to turn designers into philosophers, but to **embed critical reflection where it is most needed—at the points where AI begins to act on our behalf**.

12. Conclusion

Design thinking must evolve in the age of agentic AI. We propose Friction-Led Intelligence (FLI) as a framework that responds not just to usability, but to misalignments between reality, knowledge, and purpose. In doing so, it enables systems to adapt, reflect, and regain trust.

At its core, FLI addresses the foundational tension in agentic design:

Companies are not merely building for users—they're negotiating between their own ontological representations of bounded patterns and the user's bounded lived realities.

Friction, in this context, is not a flaw but a feature: a diagnostic signal that reveals where semantic, epistemic, or teleological drift has occurred.

In an ecosystem increasingly shaped by superplatforms and algorithmic intermediaries, design cannot afford to be neutral. It must become aware.

References

1. Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Cakmak, M., Kamar, E., & Horvitz, E. (2019). Guidelines for human–AI interaction. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM. https://doi.org/10.1145/3290605.3300233

2. Barkol, D. (2025, January 15). *Agentic AI explained: A philosophical framework for understanding AI agents*. *All Things Azure*. https://devblogs.microsoft.com/all-things-azure/agentic-philosophers/

3. Forlano, L. (2016). Decentering the human in the design of collaborative cities. *Design Issues*, 32(3), 42–54. https://doi.org/10.1162/DESI_a_00408

4. Forlano, L., & Mathew, A. (2014). From design fiction to design friction. *Journal of Urban Technology*, 21(3), 7–24. https://doi.org/10.1080/10630732.2014.942095

5. Kadel, A. (2025). *Seeds of sovereignty: Designing regenerative AI for plural epistemologies*. *SocArXiv*. https://doi.org/10.31235/osf.io/f9e65_v1

6. Kiron, D., & Schrage, M. (2019). Strategy for and with AI. *MIT Sloan Management Review*, 60(4), 44–52. https://sloanreview.mit.edu/article/strategy-for-and-with-ai/

7. Noller, J. (2024). Extended human agency: Towards a teleological account of AI. *Humanities and Social Sciences Communications*, 11, Article 1338. https://doi.org/10.1057/s41599-024-03849-x

8. Ryan, M. (2025). We're only human after all: A critique of human-centred AI. *AI & Society*, 40(3), 1303–1319. https://doi.org/10.1007/s00146-024-01976-2

9. Lin, J., Zhu, J., Zhou, Z., Xi, Y., Liu, W., Yu, Y., & Zhang, W. (2025). *Superplatforms have to attack AI agents: Epistemic friction and design governance at the platform–agent interface*. *arXiv https://arxiv.org/abs/2505.17861*

10. Zhang, A. X., Zhao, J., & Zhu, Y. (2021). Calibrating trust in human–AI collaboration. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. ACM. https://doi.org/10.1145/3411764.3445092